Administrative.

Tentative plan.

- Have all* your grades ready by Wednesday

- Review session: might feel like a recitation, I will try to send the exercises in advance.

- Recommended material:
  - past quizzes, past class, midterm homework.
  - Resources: suggested problems (midterm + final)

- Final exam: 300 points.
  - ~ 15 questions, 3hours (might be part 1,2 or 1,2,3)
  - ~ more strict: (a few examples)
    - need to see your workspace
    - camera cannot be blurred, mic has to be open
    - no headphones, you are free to mute your output.
    - might do separate conferences with screen sharing
    - scan ID on the 1st page with some work on question 1. (i.e., not the ID alone)
    - strict time limit to work on the problems.

- I won't release the final for you if the conditions are not satisfied.
- If the scan doesn't follow the rules, I will ask you to rescan it.

Lecture 12

5.3 Hamiltonian Systems
5.4 Dissipative Systems
Bifurcations

Hamiltonian Systems.

Consider the equation given on MATLAB 4:

$$2x'' = 2x - 3x^2$$

a) Write it as a system $x' = y$

$$\begin{cases} x' = y \\ y' = x - \frac{3}{2}x^2 \end{cases}$$

b) Find H s.t. $H(x,y) = c$ along solutions

$$\frac{dy}{dx} = \frac{2x - 3x^2}{2y} \Rightarrow 2y\,dy = 2x - 3x^2\,dx$$

$$y^2 = x^2 - x^3 + c$$

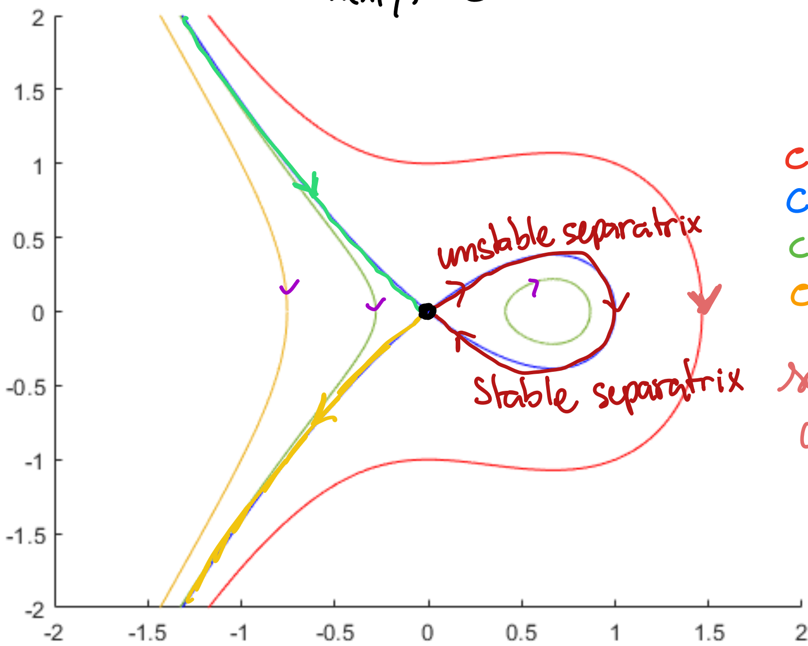
$$H(x,y) = y^2 - x^2 + x^3$$

$$\begin{cases} x' = y \\ y' = x - \frac{3}{2}x^2 \end{cases}$$

poll 3
~~a) 1~~
~~b) 2~~
~~c) 3~~
d) 4

$H(x,y) = c$

separatrices



$c=1$
$c=0$
$c=-1/10$
$c=-1$

saddle connection.

def. (saddle connection)

c) Verify that $H(x,y) = c$.

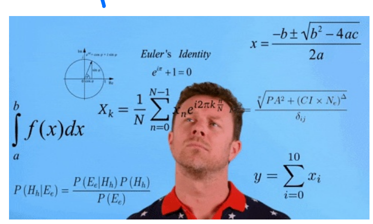
$H(x,y) = y^2 - x^2 + x^3$

$$\frac{dH}{dt} = \frac{dH}{dx}\cdot\frac{dx}{dt} + \frac{dH}{dy}\cdot\frac{dy}{dt}$$

$$= \underbrace{\left[-2x+3x^2\right]\left[y\right]} + \underbrace{\left[\cancel{2}y\right]\left[\frac{2x-3x^2}{\cancel{2}}\right]}_{-}$$

$$= 0.$$

What is the best way to verify that $f(x) = \text{constant}$?



$$\boxed{\frac{\partial H}{\partial x}}\cdot x' + \frac{\partial H}{\partial y}\cdot y' = 0$$

$-$

definition: A system of DEs is called a Hamiltonian System if there exists a real-valued function $H(x,y)$ s.t.

$$\frac{dx}{dt} = \frac{\partial H}{\partial y}$$

$$\frac{dy}{dt} = -\frac{\partial H}{\partial x}$$

H is called the Hamiltonian function for the system.

d) Verify that $H(x,y) = \text{constant}$ along solutions.

$$\frac{d}{dt}H = \frac{\partial H}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial H}{\partial y}\cdot\frac{dy}{dt}$$

$$= \frac{\partial H}{\partial x}\cdot\frac{\partial H}{\partial y} - \frac{\partial H}{\partial y}\cdot\frac{\partial H}{\partial x} = 0$$

Example: unforced. Undamped Harmonic Oscillator

$$x'' + qx = 0 \qquad \frac{dy}{dx} = \frac{y'}{x'} \iff x' \cdot \overset{dH}{dy} - y' \overset{dH}{dx} = 0$$

$$\begin{cases} x' = y \\ y' = -qx \end{cases} \qquad \frac{dy}{dx} = \frac{-qx}{y} \iff \int y\,dy = \int -qx\,dx$$

$$H(x,y) = \frac{1}{2}y^2 + \frac{q}{2}x^2 = c$$

$$\frac{\partial H}{\partial y} = y = \frac{dx}{dt}$$

$$-\frac{\partial H}{\partial x} = -qx = \frac{dy}{dt}.$$

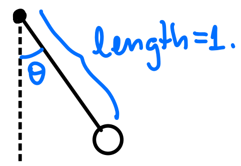
there exists $H(x,y)$ s.t.

$$x' = \frac{\partial H}{\partial y}$$

$$y' = -\frac{\partial H}{\partial x}$$

Conclusion: this is a Hamiltonian system.

Example: The ideal pendulum (Preclass)



$$-\cancel{m}g\cdot\sin\theta = \cancel{m}\cdot\ell\cdot\frac{d^2\theta}{dt^2}$$

- Introduce $v=\dot\theta$

$$\begin{cases}\dot\theta = v\\ \dot v = -\frac{g}{\ell}\sin\theta = -g\sin\theta.\end{cases}$$

$$\frac{d\theta}{dv} = \frac{v}{-g\sin\theta} \Rightarrow \int -g\sin\theta\, d\theta = \int v\, dv$$
$$c + g\cos\theta = \frac{1}{2}v^2$$

- $H(\theta, v) = \frac{1}{2}v^2 - g\cos\theta$

$v^2 = 2c + 2g\cos\theta$
$\Rightarrow v = \pm\sqrt{2c+2g\cos\theta}$

- Equilibria?

$\dot\theta = 0 \Leftrightarrow v=0$
$\dot v = 0 \Rightarrow -g\sin\theta = 0 \Rightarrow \theta = k\pi,\ k\in\mathbb{Z}$
$= \dots, -3\pi, -2\pi, -\pi, 0, \pi, 2\pi, \dots$

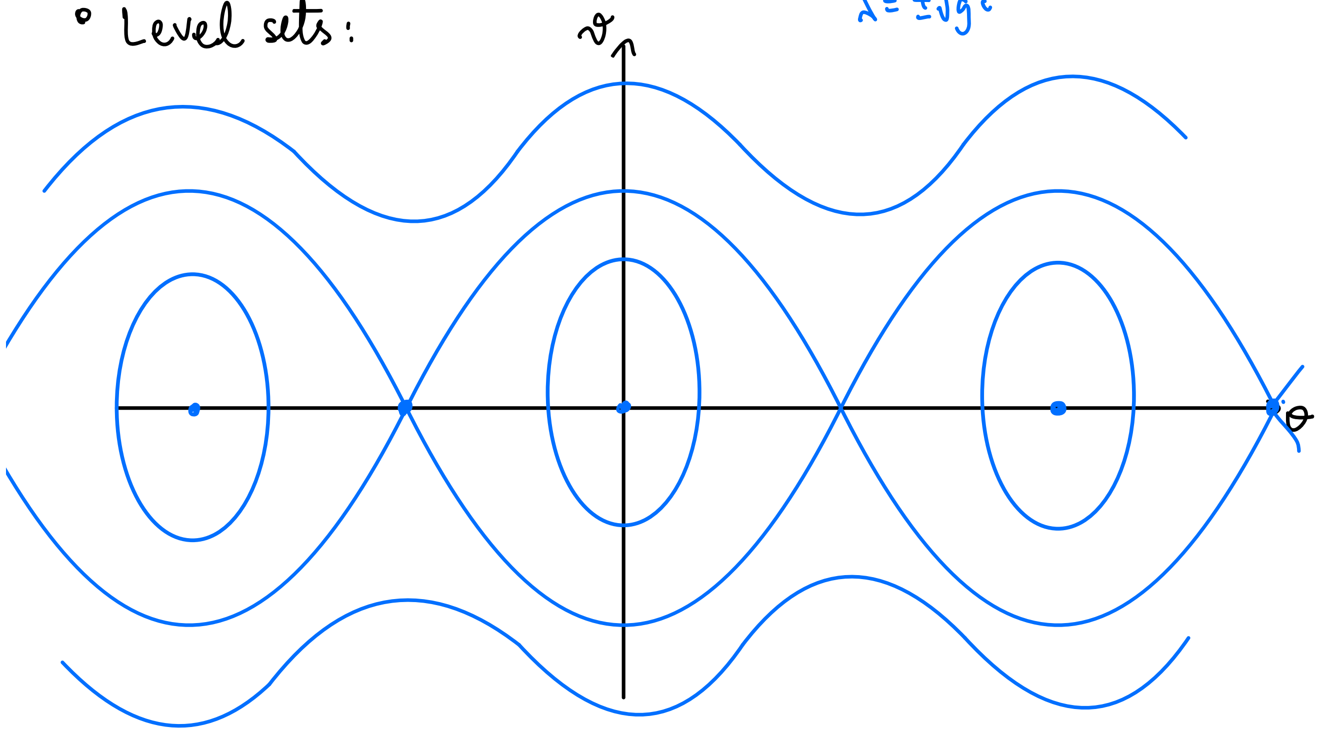
- Type of equilibria

$(0,0)$, jacobian:

$$\begin{bmatrix} 0 & 1 \\ -g\cos\theta & 0\end{bmatrix} \xrightarrow{(0,0)} \begin{bmatrix} 0 & 1 \\ -g & 0\end{bmatrix}$$

$\lambda = \pm\sqrt{g}\,i$

- Level sets:



Example: Consider the linear system $\frac{\partial f}{\partial x} = f_x$

$$x' = 42x = \frac{\partial H}{\partial y} = H_y$$
$$y' = \pi y = -\frac{\partial H}{\partial x} = -H_x$$

poll: YES / NO.

Question: Is this system Hamiltonian?

If yes, find $H(x,y)$

If no, explain why. Because if such $H(x,y)$ existed, then

$$42 = -\pi$$

which is an absurd. ⌣̈

$$\frac{\partial^2 H}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial H}{\partial y}\right) = \frac{\partial}{\partial x}(42x) = 42.$$
$$= \qquad \neq$$
$$\frac{\partial^2 H}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial H}{\partial x}\right) = \frac{\partial}{\partial y}(-\pi y) = -\pi.$$

Criterion for Hamiltonian Systems.

Suppose that

$$\frac{dx}{dt} = f(x,y) = \frac{\partial H}{\partial y}$$
$$\frac{dy}{dt} = g(x,y) = -\frac{\partial H}{\partial x}$$

is Hamiltonian.

Then $$\frac{\partial^2 H}{\partial x \partial y} = \frac{\partial f}{\partial x} = -\frac{\partial g}{\partial y} = \frac{\partial^2 H}{\partial y \partial x}$$

Ok, say $f_x = -g_y$, so what?

Goal: Find (or not) a $H(x,y)$

$$f(x,y) = \frac{\partial H}{\partial y} \Rightarrow H(x,y) = \int f(x,y)\,dy + \varphi(x)$$

$$\underbrace{\frac{\partial H}{\partial x}}_{-g(x,y)} = \frac{\partial}{\partial x}\int f(x,y)\,dy + \varphi'(x) \Rightarrow \varphi'(x) = -\frac{\partial}{\partial x}\int f(x,y)\,dy - g(x,y)$$

$$\varphi(x) = \int -\frac{\partial}{\partial x}\int f(x,y)\,dy - g(x,y)\,dx. + c_1$$

$H(x,y) = c$

$$H(x,y) = \int f(x,y)\,dy - \int\left[\frac{\partial}{\partial x}\int f(x,y)\,dy\right] + g(x,y)\,dx$$

if Hamiltonian $\Rightarrow f_x = -g_y \Rightarrow$ Hamiltonian

Ex: Consider

$$\frac{dx}{dt} = -x\sin y + 2y \quad = f(x,y)$$

$$\frac{dy}{dt} = -\cos y \quad = g(x,y)$$

Determine if the system is Hamiltonian, and if so, find H.

- Q: is $f_x = -g_y$?

$$f_x = -\sin y$$

$$= -$$

$$g_y = \sin y$$

$$f(x,y) = \frac{\partial H}{\partial y} = -x\sin y + 2y \Rightarrow H(x,y) = \int -x\sin y + 2y \, dy + \varphi(x)$$

$$= x\cos y + y^2 + \varphi(x)$$

$$\frac{\partial H}{\partial x} = \cos y + \varphi'(x) = -g(x,y) = \cos y \Rightarrow \varphi'(x) = 0 \Rightarrow \varphi(x) = \text{const.}$$

$$= 0.$$

$$H(x,y) = x\cos y + y^2.$$

7:49. break.

Equilibrium Points of Hamiltonian Systems.

Suppose (x_0, y_0) is an equilibrium point.

$$\begin{cases} \dfrac{dx}{dt} = \dfrac{\partial H}{\partial y} = f(x,y) \\[2ex] \dfrac{dy}{dt} = -\dfrac{\partial H}{\partial x} = g(x,y) \end{cases}$$

$$\begin{bmatrix} \dfrac{\partial^2 H}{\partial x \partial y} & \dfrac{\partial^2 H}{\partial y^2} \\[2ex] -\dfrac{\partial^2 H}{\partial x^2} & -\dfrac{\partial^2 H}{\partial x \partial y} \end{bmatrix} = \begin{bmatrix} f_x & f_y \\[2ex] g_x & g_y \end{bmatrix}$$

evaluated at (x_0, y_0)

$$\det\left(\begin{bmatrix} a & b \\ -c & -a \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0 \quad ; \quad (a-\lambda)\cdot(-a-\lambda) + bc = 0$$

$$\lambda^2 - a^2 + bc = 0$$

$$\lambda = \pm\sqrt{a^2 - bc}$$

Case 1: $a^2 - bc < 0 \Rightarrow \lambda = \pm ki.$

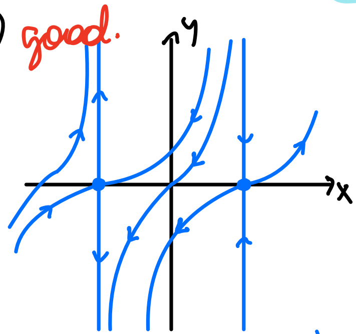
Case 2: $a^2 - bc = 0 \Rightarrow \lambda = 0$

inconclusive since $Re(\lambda) = 0$.

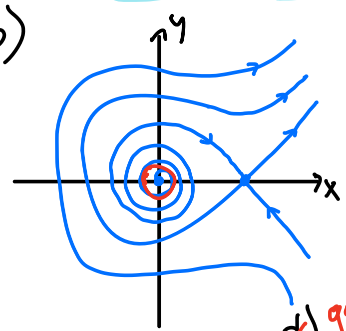
Case 3: $a^2 - bc > 0 \Rightarrow \lambda = \pm k$ saddle

Ex. Which of the following phase portraits could possibly be the phase portrait of a Hamiltonian system?

~~a~~) good.

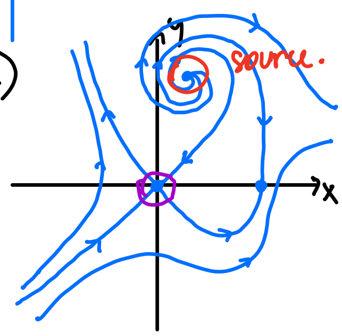


b)

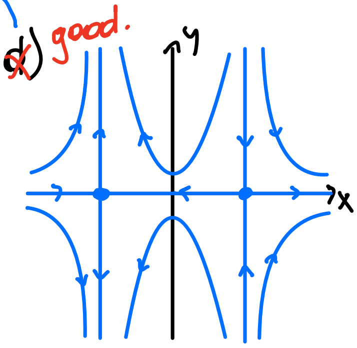


source

c)



~~d~~) good.

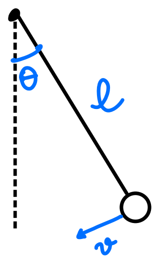


Sol. a, d might be

b, c nope.

5.4. Dissipative Systems

The not so perfect pendulum



$$\ddot{\theta} + \frac{b}{m}\dot{\theta} + g/\ell \sin\theta = 0$$

Introduce $v = \dot{\theta}$

$$\dot{\theta} = v \qquad = 0 \Rightarrow v = 0$$

$$\dot{v} = -g/\ell \sin\theta - \frac{b}{m} v \quad = 0 \overset{v=0}{\Rightarrow} -\frac{g}{\ell}\sin\theta = 0 \Rightarrow \theta = k\pi,\ k \in \mathbb{Z}$$

$$= \ldots, -2\pi, -\pi, 0, \pi, 2\pi, \ldots$$

Equilibrium points:

$(k\pi, 0),\ k \in \mathbb{Z}$

Jacobian matrix of the vector field:

$$J = \begin{bmatrix} 0 & 1 \\ -\frac{g}{\ell}\cos\theta & -\frac{b}{m} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ -g/\ell & -b/m \end{bmatrix} \ldots (-4\pi, 0), (-2\pi, 0), (0, 0), (2\pi, 0), (4\pi, 0) \ldots$$

$$\begin{bmatrix} 0 & 1 \\ g/\ell & -b/m \end{bmatrix} \ldots (-5\pi, 0), (-3\pi, 0), (-\pi, 0), (\pi, 0), (3\pi, 0), (5\pi, 0) \ldots$$

if $(0,0), (\pm 2\pi, 0), (\pm 4\pi, 0), \ldots$

$$J = \begin{bmatrix} 0 & 1 \\ -g/\ell & -b/m \end{bmatrix}$$

$(-\lambda)(-\lambda - \frac{b}{m}) + g/\ell = 0$

$\lambda^2 + \frac{b}{m}\lambda + g/\ell = 0$

$\lambda = \frac{-b}{2m} \pm \underbrace{\overbrace{\sqrt{\left(\frac{b}{2m}\right)^2 - g/\ell}}^{=\text{Real}}}_{\text{imaginary}} = 0$ positive

$tr\, J = b/m$

$\det J = g/\ell$

$$Re(\lambda) < 0$$

then $Re(\lambda) = \frac{-b}{2m} < 0$

$Re(\lambda) =$

$Re(\lambda) \quad < 0$

$B^2 - C \ , \ C > 0$

$B^2 - C < B^2$

$-B \pm \sqrt{B^2 - C} < -B \pm \sqrt{B^2} \leq 0.$

$(2k\pi, 0)$ always have $Re(\lambda) < 0$ sink or spiral sink.

if $(\pm\pi, 0), (\pm 3\pi, 0), (\pm 5\pi, 0), \ldots$

$$J = \begin{bmatrix} 0 & 1 \\ g/\ell & -b/m \end{bmatrix}$$

$\operatorname{tr} J = -b/m$

$\det J = -g/\ell < 0$

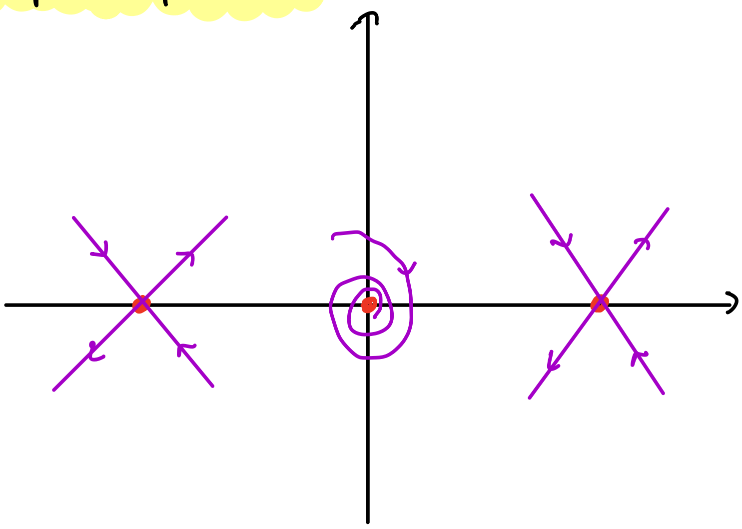
$(-\lambda)(-\lambda - b/m) - g/\ell = 0$

$\lambda^2 + \frac{b}{m}\lambda - g/\ell = 0$

$\lambda = \frac{-b}{2m} \pm \sqrt{\underbrace{\left(\frac{b}{2m}\right)^2 + g/\ell}_{=0?\ \text{never!}}}$

$(2k\pi + \pi, 0)$ are saddles

Draw the phase portrait:



What about our good friend Hamilton?

$$H(\theta, v) = \frac{1}{2}v^2 - \frac{g}{\ell}\cos\theta.$$

$$\frac{dH}{dt} = \frac{\partial H}{\partial \theta}\cdot\frac{d\theta}{dt} + \frac{\partial H}{\partial v}\cdot\frac{dv}{dt}$$

$$= \left[\frac{g}{\ell}\cdot\sin\theta\right]\left[v\right] + \left[v\right]\left[-\frac{b}{m}v - \frac{g}{\ell}\sin\theta\right]$$

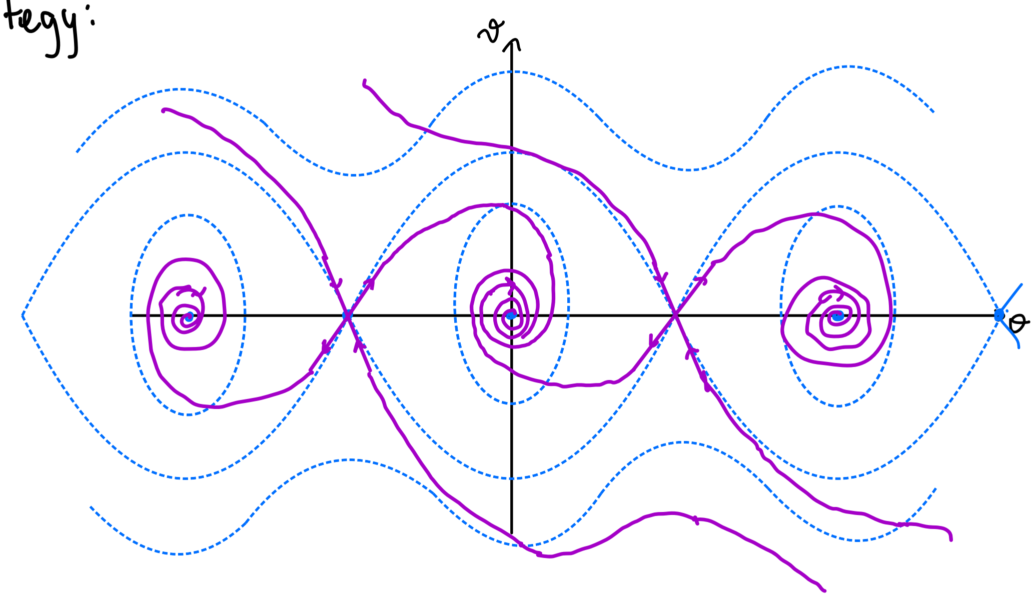
$$= \cancel{v\frac{g}{\ell}\sin\theta} - \frac{b}{m}v^2 - \cancel{v\frac{g}{\ell}\sin\theta}.$$

$$= -\frac{b}{m}v^2 \leq 0$$

$b>0$
$m>0$
$v^2 \geq 0$

Conclusion: H is decreasing along solutions of the system.

Strategy:



What did we use?

def. A function $L(x,y)$ is called **Lyapunov function** for a system of diff. eq. if for every solution $(x(t), y(t))$

$$\frac{d}{dt} L(x(t), y(t)) \leq 0$$

Ex Damped Harmonic Oscillator: $x'' + px' + qx = 0$

$$\begin{cases} x' = y \\ y' = -qx - py \end{cases}$$

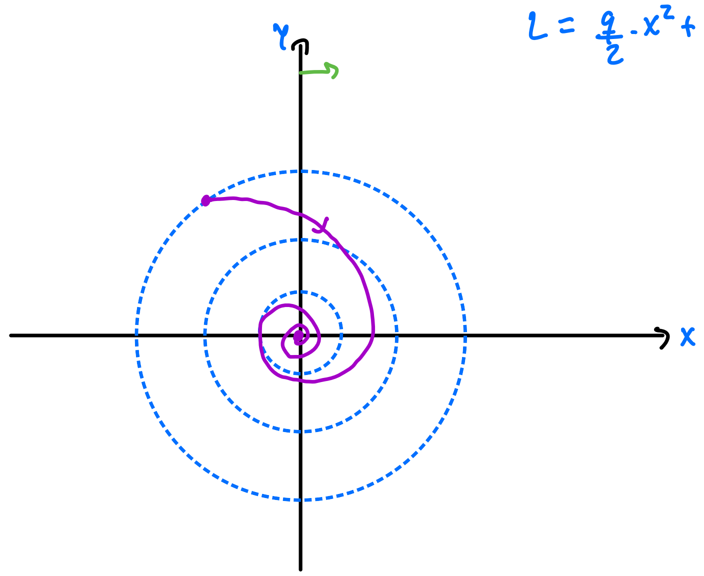
$$L(x,y) = \frac{q}{2}x^2 + \frac{1}{2}y^2$$

homework before going to bed.

- Check if $\frac{dL}{dt} \leq 0$

- Sketch the phase portrait

$L = \frac{q}{2} \cdot x^2 + \frac{1}{2}y^2$

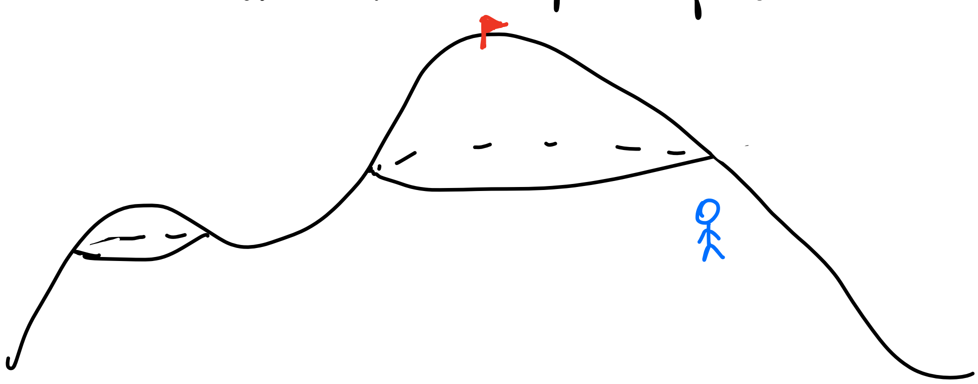


Gradient Systems.

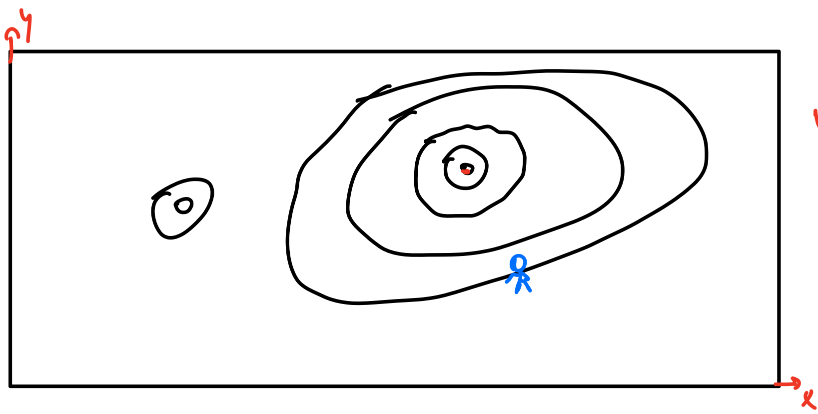
Claim: Finding a Lyapunov function is not that easy.

but sometimes it is.

Suppose you want a good grade in this course and to get that you need to climb to the peak of a mountain:



Unfortunately, you are blind and can only check which direction the altitude increases the fastest.



If $z = G(x,y)$ is the altitude, what is the direction?

$$\nabla G = \left(\frac{\partial G}{\partial x}, \frac{\partial G}{\partial y} \right)$$

And if you had a vector for each point?

VECTOR FIELD!

def A system of diff. eq. is called a gradient system if there is a function G s.t.

$$\begin{cases} x' = \dfrac{\partial G}{\partial x} \\[2ex] y' = \dfrac{\partial G}{\partial y} \end{cases}$$

Exercise: Compute $\dfrac{dG}{dt}$.

$$\begin{aligned} \frac{dG}{dt} &= \frac{\partial G}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial G}{\partial y} \cdot \frac{dy}{dt} \\ &= \frac{\partial G}{\partial x} \cdot \left[\frac{\partial G}{\partial x}\right] + \frac{\partial G}{\partial y} \cdot \left[\frac{\partial G}{\partial y}\right] \\ &= [G_x]^2 + [G_y]^2 \geq 0 \end{aligned}$$

$L = -G$ is a Lyapunov function for this kind of system.

(so $-G$ decreases along solutions)

Equilibrium Points in Gradient Systems

(x_0, y_0)

$$x' = \frac{\partial G}{\partial x} \qquad A = \begin{bmatrix} G_{xx} & G_{xy} \\ G_{yx} & G_{yy} \end{bmatrix} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

$$y' = \frac{\partial G}{\partial y}$$

$$\det(A - \lambda I) = (a-\lambda)(c-\lambda) - b^2 = 0$$

$$\lambda^2 - (a+c)\lambda - b^2 + ac = 0$$

$$\lambda = \frac{a+c}{2} \pm \frac{1}{2}\sqrt{(a+c)^2 + 4b^2 - 4ac}$$

$$a^2+2ac+c^2 - 4ac = a^2 - 2ac + c^2$$

$$= \frac{a+c}{2} \pm \frac{1}{2}\sqrt{\underbrace{(a-c)^2 + 4b^2}_{\geq 0}}$$

Conclusion: the eigenvalues are always real.

i.e., gradient systems DO NOT have sp. source, sp. sink or center.

<u>Question:</u> is the converse true?

if there is a Lyapunov function, is it a Gradient system? ~~Poll~~

No.

Ex: $\begin{cases} x' = -x+y \\ y' = -x-y \end{cases}$ $\qquad L(x,y) = x^2+y^2.$

a) Verify that L is a Lyapunov function.

$$\frac{dL}{dt} = L_x \cdot x' + L_y \cdot y' = (2x)\cdot(-x+y) + (2y)(-x-y)$$
$$= -2x^2 + \cancel{2xy} - \cancel{2xy} - 2y^2 = -(2x^2+2y^2) \leq 0.$$

b) Classify the equilibria.

$A = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \Rightarrow \lambda^2 + 2\lambda + 2 = 0$

$\lambda = \frac{-2}{2} \pm \frac{1}{2}\sqrt{4-8} = -1 \pm i \Rightarrow$ spiral sink.

$= \sqrt{-4} = 2i$

c) Put 2 and 2 together.

Lyapunov ☑ $\qquad$ Type of equilibria that cannot happen in Gr. systems ☑


Dobby is free